MUHAMMAD HILAL DARUL FAUZAN

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Final year computer science student at the Universitas Indonesia with hands-on experience in software engineering and AI/ML projects, and proven leadership roles in multiple organizations. Proficient in software engineering tools, passionate about crafting innovative design solutions, and eager to gain valuable experience. I am passionate about designing and building scalable, robust software systems that integrate analytical insights from data. With a strong foundation in software development and AI/ML, I aspire to develop tools and applications that optimize processes, enhance decision-making, and improve lives by combining solid engineering practices with technical expertise.

EDUCATIONS

Universitas Indonesia - Depok, Jawa Barat, Indonesia — August 2022 - August 2026

Bachelor of Computer Science | CGPA: 3.65/4.00

- Relevant Courseworks: Software Engineering, AI & Data Science, Knowledge Graph, Computer Vision, Deep Learning, Web Design & Development, Data Structures & Algorithms

SMA Negeri 1 Jakarta - Jakarta Pusat, Jakarta, Indonesia — July 2019 - May 2022

Mathematic and Natural Science | Grade: 90.57/100

- Participated in the National Science Olympiad in Astronomy at the provincial-level (Jakarta) and ranked in 6th place
- Became a top 10 outstanding student in the school

WORK EXPERIENCES

GDP Labs - Jakarta Selatan, Jakarta, Indonesia

AI Engineer Intern — March 2026 - Present

- Led GLChat's guardrail migration to the in-house GenAI SDK, replacing rule-based phrase matching with LLM-based classification across 16 content-safety categories with configurable enforcement modes and topic controls, improving multilingual (Indonesian/English) accuracy and reducing false positives by ~30%
- Extended the SDK and its integration with conversation-history-aware decisions, additional safety categories, and admin-configurable controls including a provider-agnostic model and a conservative fail-safe mode, plus per-preset caching for lower latency
- Shipped runtime configuration for External Pipeline Services, a FastAPI service exposing Beebot's chat pipelines as OpenAI Responses API-compatible endpoints, letting admins update pipeline config through validated APIs that persist to PostgreSQL and propagate across replicas via Redis with zero service restart
- Conducted guardrail research explorations across KB and KG grounding, streaming-output validation, factual grounding, and built a 175-case bilingual evaluation dataset labeled by harm category, severity, and evasion technique

PT Bringin Inti Teknologi (bitcorp.) - Jakarta Pusat, Jakarta, Indonesia

Software Engineer Intern — July 2025 - January 2026

- Designed and delivered end-to-end workforce management platform with mobile and web interfaces, enabling efficient employee tracking, and project management
- Developed a facial recognition attendance system with face embedding extraction, anti-spoofing detection, and GPS validation, impersonation in attendance and ensuring 95%+ identity verification accuracy to prevent fraudulent clock-ins
- Implemented automated approval systems for leave, overtime, and timesheets with real-time notification channels and analytics dashboards, decreasing manual processing time from 3 days to 4 hours and improving team productivity by 60%

Universitas Indonesia - Depok, Jawa Barat, Indonesia

Teaching Assistant — August 2024 - October 2025

- Tutored 250+ undergraduate students in Computer Vision, Statistics & Probability, and Calculus 2 courses
- Graded homework assignments, quizzes, and coding projects with clear feedback across all three subjects
- Helped students connect theory to practical applications in mathematics and programming
- Helped in preparing course materials, including practice problems for exam preparation

HONORS & AWARDS

RSA PTI Scholarship Awardee (2025)
International Youth Innovation Summit #15 - Chapter Malaysia, Singapore, Thailand (2025): The Most Creative Delegate and 2nd Best Video Project Innovation
Kompetisi Sains Nasional (2020): 6th Place in the Provincial-level National Science Olympiad in Astronomy
Primagama Mencari Juara (2017): 1st Place and Gold Medalist in Mathematics, Provincial-level Jakarta

PROJECTS

MAAMS

- A NextJS and Django website app for a software engineering project course with a team of 7 individuals, a root cause validator using DeepSeek R1 model
- Served as a full-stack developer for 4 sprints, managing action items, backlogs, acceptance criteria, and developed authentication and validator features
- Developed with Scrum in collaboration with FISIP UI and supported the Philosophy of Communication course by analyzing and confirming communication-related problems

Apap Medika

- Developed a hospital management system featuring 6 microservices: Profile, Appointment, Hospitalization, Pharmacy, Insurance, and Bill services
- Designed the entire app as a UI designer, created the design system from scratch, and defined user flows
- Developed complex business including appointment scheduling, reservations, pharmacy inventory, insurance policy management, and integrated billing with payment tracking

LembarPena - Best Web Application Award of Platform Based Programming Course 2023

- A web and mobile application project with a literacy community theme, created using Django and Flutter
- Connected readers and authors, offering book purchasing, discussion forums, and a wishlist feature
- Designed the entire app as a UI designer, created the design system from scratch, and defined user flows

CrawlKit API

- Built a REST API for intelligent web scraping and data extraction, featuring robots.txt compliance, concurrent crawling, and real-time analytics dashboard
- The system used BeautifulSoup for HTML parsing with advanced CSS selectors
- Implemented ethical crawling practices with automatic delays and rate limiting, provided reporting capabilities for data quality monitoring

Database Backup Utility

- Built a CLI tool in Go for automated database backup/restore supporting MySQL, PostgreSQL, MongoDB, and SQLite
- Implemented multi-cloud storage (AWS S3, GCS, Azure) with compression, encryption, and retention policies

Sentiment Analysis: DC Universe Rebrand (Man of Steel vs. Superman 2025)

- Designed and executed a full-cycle research project to evaluate the shift in public sentiment regarding the DC Universe's creative direction
- The study compared audience reception between Zack Snyder's Man of Steel (2013) and James Gunn's Superman (2025) using social media data

Natural Language Inference

- Developed a deep learning model for NLI to classify relationships between premise and hypothesis as entailment, contradiction, or neutral
- Used custom-built architectures with the pre-trained DeBERTa V3 model in PyTorch, trained from scratch on Indonesian and English data

Mario Object Detection

- Developed an object detection system using YOLO v11 to accurately and efficiently detect Mario characters in video datasets
- Addressed real-world challenges like lighting variability, angles, and complex backgrounds
- The system was evaluated on accuracy and processing speed, providing insights into implementing and optimizing YOLO for dynamic visual data

Health Admission Prediction

- Conducted exploratory data analysis to identify patient traits and disease trends, including peak admission periods and disease patterns by age and smoking status
- Built classification models to predict patient outcomes and regression models to estimate ICU stay duration
- Developed clustering models to group patients, aiming to support medical decisions, optimize hospital resources, and enhance healthcare quality

SKILLS AND TOOLS

Programming Languages: Python, TypeScript, Java, Go
AI/ML: PyTorch, TensorFlow, LangChain, LangGraph, NeMo Guardrails, N8N, OpenCV, YOLO, Hugging Face Transformers
Backend: FastAPI, Django, NestJS, Spring Boot
Frontend/Mobile: Next.js, React.js, Vue.js, Tailwind CSS, Flutter
Databases: PostgreSQL, MySQL, MongoDB
Cloud & Tools: Docker, Google Cloud, AWS